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10 / 10 submitted

Q1. Quadratic equation

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>
#include <math.h>

int main() {
    float a, b, c,d,root1,root2;
    if (scanf("%f%f%f", &a, &b, &c)!=3) return 0;
    d=b*b-4*a*c;
    if(d>=0){
        root1 = (-b+sqrt(d))/(2*a);
        root2 = (-b-sqrt(d))/(2*a);
        printf("%.2f %.2f\n",root1,root2);
    }
    return 0;
}
```

OUTPUT

```
Compilation Error:
C:\Windows\TEMP\sandbox_192312284RD_6a61a136063dd8.16537494\main.c: In function 'main':
C:\Windows\TEMP\sandbox_192312284RD_6a61a136063dd8.16537494\main.c:6:41: error: 'rturn' undeclared (first use in this function)
   6 |         if (scanf("%f%f%f", &a, &b, &c)!=3) rturn 0;
     |                                         ^~~~~~
C:\Windows\TEMP\sandbox_192312284RD_6a61a136063dd8.16537494\main.c:6:41: note: each undeclared identifier is reported only once for each function it appears in
C:\Windows\TEMP\sandbox_192312284RD_6a61a136063dd8.16537494\main.c:6:46: error: expected ';' before numeric constant
   6 |         if (scanf("%f%f%f", &a, &b, &c)!=3) rturn 0;
     |                                         ^~
     |                                         ;
```

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Q2. Prime numbers

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
int main(){
    int n,i,j,prime;
    int first_number=1;
    if(scanf("%d",&n)!=1){
        return 0;
    }
    if(n<=1){
        printf("No prime numbers");
        return 0;
    }
    for(i=2;i<=n;i++){
        prime = 1;
        for(j=2;j*j<=i;j++){
            if(i%j==0){
                prime=0;
                break;
            }
        }
        if(prime==1){
            if(!first_number){
                printf(" ");
            }
            printf("%d",i);
            first_number =0;
        }
    }
    printf("\n");
    return 0;
}
```

INPUT

20

OUTPUT

2 3 5 7 11 13 17 19

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Q3. Palindrome

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i, len, flag = 1;

    if (scanf("%s", str) != 1) {
        return 0;
    }

    len = strlen(str);

    for (i = 0; i < len / 2; i++) {
        if (str[i] != str[len - i - 1]) {
            flag = 0;
            break;
        }
    }

    if (flag) {
        printf("Palindrome\n");
    } else {
        printf("Not a Palindrome\n");
    }

    return 0;
}
```

INPUT

OUTPUT

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Q4. Pyramid

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    int n, i, j, space;

    if (scanf("%d", &n) != 1) {
        return 0;
    }

    for (i = 1; i ≤ n; i++) {
        for(space=1; space ≤ n-i; space++){
            printf(" ");
        }
        for (j = 1; j ≤ i; j++) {
            printf("%d", j);
        }
        for(j=i-1; j ≥ 1; j--){
            printf("%d", j);
        }
        printf("\n");
    }

    return 0;
}
```

INPUT**OUTPUT**

```
1
121
12321
1234321
123454321
```

Q5. Voting

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>

int main() {
    int age;
    scanf("%d", &age);

    if (age ≥ 18) {
        printf("Eligible");
    }
    else {
        printf("Not eligible");
    }

    return 0;
}
```

INPUT**OUTPUT**

Q6. Calculator

Score: 3.33 TC: 1/3 passed

CODE

```
#include <stdio.h>

int main() {
    char op;
    int num1, num2;

    if (scanf("%d %c %d", &num1,&op,&num2) == 3) {
        switch (op) {
            case '+':
                printf("Add result = %d", num1 + num2);
                break;
            case '-':
                printf("Subtraction result = %d", num1 - num2);
                break;
            case '*':
            case 'x':
            case 'X':
                printf("Output: Multiplication result = %d", num1 * num2);
                break;
            case '/':
                if (num2 != 0.0) {
                    printf("Division result = %d", num1 / num2);
                } else {
                    printf("Error: Division by zero");
                }
                break;
            default:
                printf("Error: Invalid operator");
        }
    } else {
        printf("Error: Invalid input format");
    }

    return 0;
}
```

INPUT

4 * 3

OUTPUT

Output: Multiplication result = 12

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Q7. Month counting

Score: 0.00 TC: 0/2 passed

CODE

```
#include <stdio.h>
int main() {
    int month;
    scanf("%d", &month);
    printf("Input month number:%d\n",month);
    switch (month) {
        case 1:
        case 3:
        case 5:
        case 7:
        case 8:
        case 10:
        case 12:
            printf("31\n");
            break;
        case 4:
        case 6:
        case 9:
        case 11:
            printf("30\n");
            break;
        case 2:
            printf("28\n");
            break;
        default:
            printf("Invalid month\n");
            break;
    }
    return 0;
}
```

INPUT

7

OUTPUT

Input month number:7
31

Q8. Positive or negative

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>
int main(){
    int a;
    scanf("%d",&a);
    if(a>0){
        printf("Positive\n");
    }
    else if(a<0){
        printf("Negative\n");
    }
    else{
        printf("Zero\n");
    }
    return 0;
}
```

INPUT

-1

OUTPUT

Negative

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Q9. Multiplication Table

Score: 10.00 TC: 2/2 passed

CODE

```
#include <stdio.h>
int main(){
    int n,i;
    if(scanf("%d",&n)≠1){
        return 0;
    }
    for(i=1;i≤10;i++){
        printf("%d x %d = %d\n",n,i,n*i);
    }
    return 0;
}
```

INPUT

5

OUTPUT

```
Enter a number:
5 X 1 = 5
5 X 2 = 10
5 X 3 = 15
5 X 4 = 20
5 X 5 = 25
5 X 6 = 30
5 X 7 = 35
5 X 8 = 40
5 X 9 = 45
5 X 10 = 50
```

Q10. Factorial

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
int main(){
    int n,i;
    long long fact = 1;
    if(scanf("%d",&n)≠1){
        return 0;
    }
    for (i=1;i≤n;i++){
        fact*=i;
    }
    printf("%lld\n",fact);
    return 0;
}
```

INPUT

5

OUTPUT

120